

Go Business

Referral Dashboard — Coding Assessment

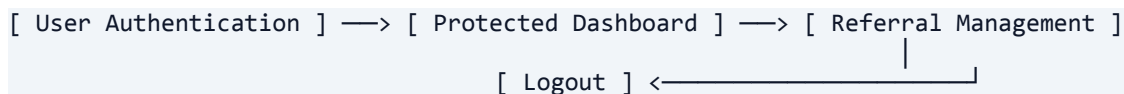
1. Application Overview

This document outlines the requirements and flow for the Referral Dashboard Application, a web-based productivity tool built for Go Business to help users track referrals, earnings, and partner activity through a structured dashboard.

Objective

To build a secure, responsive, and intuitive referral management system featuring user authentication, a referrals overview pulled from an API, a searchable, sortable, and paginated referrals table, referral link and code sharing, and a referral detail page.

2. Application Flow



1. User Authentication: Users must log in via a secure endpoint using their email and password.
2. Dashboard Access: Upon successful authentication, users are redirected to the protected referral dashboard at /.
3. Referral Management: Users can view overview metrics, service summary details, their referral link/code, and search and sort their referrals via the API, with the results paginated client-side at 10 rows per page.
4. Logout: Users can securely log out, which clears the session cookie and returns them to the login screen.

Routing Notes

- Unauthenticated users attempting to access a protected route (/ or /referral/:id) are redirected to /login.
- Authenticated users visiting /login are redirected to /.
- The Not Found route (*) must remain reachable without a token and must not be wrapped in ProtectedRoute.
- App.jsx must wrap <Routes> in <BrowserRouter> so the whole app works when the portal only loads App.jsx; main.jsx should render <App /> only, with no router there.

3. Authentication API

Login Endpoint

Method: POST

URL: <https://v9fes04dwf.execute-api.eu-north-1.amazonaws.com/api/auth/signin>

Request Payload

```
{
  "email": "<user_email>",
  "password": "<user_password>"
}
```

Success Response

```
{
  "data": {
    "token": "<jwt_token_string>"
  }
}
```

What happens on success: The token is read from `responseJson.data.token` (nested under `data`) and saved with `Cookies.set('jwt_token', token)`. The application then navigates the user to `/`.

Failure Response (401)

```
{
  "message": "Invalid email or password"
}
```

What happens on failure: The message is read from `responseJson.message` and the error text "Invalid email or password" is displayed on the login page. The Sign in button stays enabled even when fields are empty — every click performs the POST to the login API, so empty fields, partial fields (email only or password only), or wrong credentials all receive the API error response and the user remains on `/login`.

4. Route Structure & Page Layouts

Route Structure

Route	Type	Notes
<code>/login</code>	Public	Authenticated users are redirected to <code>/</code>
<code>/</code>	Protected	Requires <code>jwt_token</code> cookie; the Referral Dashboard
<code>/referral/:id</code>	Protected	Detailed view of an individual referral
<code>/dashboard/referrals</code>	Optional	May redirect to <code>/</code> , same dashboard
<code>*</code>	Public	Not Found / 404 page; must NOT be wrapped in <code>ProtectedRoute</code>

Login Page (`/login`)

UI Elements:

- Brand title: Go Business
- Tagline: *"Sign in to open your referral dashboard."* (including the trailing period)
- Email input field (placeholder="you@example.com"), associated label "Email" with `htmlFor/id` wired correctly for accessibility
- Password input field (type="password"), associated label "Password"
- "Sign in" submit button (exact label text)

Validation Rules:

- Sign in remains enabled regardless of field contents; the POST request fires on every click, and the API's own error response governs whether the user proceeds or stays on the page.

Navbar (Dashboard & Detail pages)

- Brand link text "Go Business", linking to / (recommended for accessibility: add aria-label="Go to dashboard home")
- A nav containing at least a Home link (recommended for accessibility: add aria-label="Primary")
- "Log out" button (exact text) that clears the jwt_token cookie and navigates to /login

Dashboard Page (/)

Header:

- Main heading (<h1> or equivalent): Referral Dashboard
- Subtitle paragraph: "Track your referrals, earnings, and partner activity in one place."

Loading & Error States:

- While data is loading, a loading state may be shown. If the referrals GET rejects or returns a non-success HTTP status, an error message is displayed, surfacing whatever the client knows about the failure — typically the message string from a JSON error body together with the status code when the backend includes one. (Recommended for accessibility: mark this region with role="alert" so screen readers announce it automatically.)

Overview Section:

- Section title heading: Overview
- Renders each metric from the API with its label and value (order is flexible; values must appear in the document text)
- Recommended for accessibility: wrap the section in a landmark role="region" with aria-label="Overview metrics"

Service Summary Section:

- Section title: Service summary
- Labels (exact copy): Service, Your Referrals, Active Referrals, Total Ref. Earnings
- Values sourced from serviceSummary in the API response
- Recommended for accessibility: add aria-label="Service summary" on the section (or equivalent region)

Share Referral Section:

- Panel title: "Refer friends and earn more"
- Labels: Your Referral Link, Your Referral Code
- Read-only inputs (or visible text) showing referral.link and referral.code
- Two buttons with exact text "Copy" (one for the link, one for the code)
- Recommended for accessibility: add aria-label="Share referral" on the section

All Referrals Table:

Section title: All referrals

Column	Description	Empty State
Name	Referred partner's name	No matching entries

Column	Description	Empty State
Service	Service name associated with the referral	No matching entries
Date	Referral date, displayed as YYYY/MM/DD (converted from the API's ISO YYYY-MM-DD)	No matching entries
Profit	Referral profit, formatted as USD currency with no decimal digits, en-US style (e.g. \$1,234)	No matching entries

- Search input: placeholder="Name or service..." (ellipsis character). Typing here triggers a new API call (see Section 6 — Referrals API Logic) that returns only matching rows by name or service; the table re-renders from that response. (Recommended for accessibility: add aria-label="Search referrals")
- Sort control: a select wrapped in a label whose visible text includes "Sort by date"; options include "Newest first" (value="desc") and "Oldest first" (value="asc"); default is newest first (desc). Changing this triggers a new API call that returns the rows already sorted by date.
- Pagination: handled entirely on the client. The API does not paginate — it returns the full (filtered/sorted) list for a given search or sort request. The app must slice that list into pages of 10 rows each and render "Previous" and "Next" buttons (exact labels); Previous disabled on page 1; Next disabled on the last page; numbered page buttons shown when there are multiple pages.
- Footer summary text pattern: "Showing <from>–<to> of <total> entries" using an en dash (–) between the numbers, e.g. "Showing 1–10 of 50 entries"
- Row click (or activation) navigates to /referral/<id>

Footer (Dashboard)

- A <footer> element
- Brand text "Go Business" (exact; may appear alongside copyright)
- A nav including links "About" and "Privacy" (href placeholders are fine; recommended for accessibility: add aria-label="Footer")
- Copyright line containing "© 2024 Go Business"

Referral Details Page (/referral/:id)

Accessible by clicking on a specific row in the referrals table.

- Loaded via GET on the referrals URL with ?id=<id> using the Bearer token; if the parsed body does not resolve to a row for that id (wrong URL, missing param, or unexpected JSON shape), the page falls back to the not-found experience below.
- For a single referral fetch, the data field is often the row itself — an object with id, name, serviceName, date, and profit — rather than only an entry nested inside a referrals array. This counts as a successful match when the ids line up, so deep links populate the detail view instead of falling through to not-found.

When data is loaded, displays:

- A main heading: Referral Details
- The partner's name as a heading
- Definition list (or equivalent) rows including labels Referral ID, Service Name, plus date and profit using the same display formatting as the dashboard table

- A link whose accessible name matches "Back to dashboard" (link text may include a leading arrow, e.g. ← Back to dashboard)

If the API returns no row / 404, displays heading: "Referral not found"

5. Referral Component Requirements

Overview Metric Requirements

Each metric rendered from the API must display:

- Label (from the metrics array)
- Value (from the metrics array)

Referral Table Row Requirements

Every row within the All Referrals table must display:

- Name
- Service
- Date (formatted YYYY/MM/DD)
- Profit (formatted as USD, no decimal digits)

Share Referral Fields

- Your Referral Link (read-only, with Copy button)
- Your Referral Code (read-only, with Copy button)

Not Found Page

- Displays 404 and Page not found
- Includes a Back to dashboard link that navigates to /

6. Technical Implementation Details

Cookie Implementation

- Auth Token: Stored cookie name: `jwt_token` (set via `Cookies.set('jwt_token', token)` on login success; removed on logout)

Protected Route Logic

- `ProtectedRoute` checks for the cookie using `Cookies.get('jwt_token')`
- If the cookie exists, render the component
- If the cookie is missing, redirect automatically to `/login`
- The `*` (Not Found) route must not be wrapped in `ProtectedRoute`

Authentication Flow (How the Token Works)

There is no separate, fixed API key for the dashboard or referrals data. The only credential the candidate needs is the test email and password from Section 7, used once against the login endpoint.

- Step 1 — Sign in: POST the email/password to the login endpoint (Section 3). The response contains a JWT at `data.token`.
- Step 2 — Store the token: Save that JWT in the `jwt_token` cookie (`Cookies.set('jwt_token', token)`). This token is the only credential used for every subsequent request.

- Step 3 — Reuse the token: For both the referrals list and referral detail requests, read the token back out of the cookie (Cookies.get('jwt_token')) and send it as Authorization: Bearer <jwt_token>. There is nothing else to configure — no API key, no separate secret.
- If the token is missing or expired, the referrals API will reject the request; in that case, treat it the same as any other failed fetch (show the error message/status, per the Dashboard Page loading & error behavior above).

Referrals API Logic

All requests below use the same base URL with different query parameters, and all require the Bearer token described above.

Purpose	Request
Full list (default)	GET /api/referrals
Search by name or service	GET /api/referrals?search=pm (also accepts ?q=graphics)
Single referral by id	GET /api/referrals?id=5
Sort by date	GET /api/referrals?sort=asc (asc = oldest first, desc = newest first; default is desc)

- Base URL for all of the above:
<https://v9fes04dwf.execute-api.eu-north-1.amazonaws.com/api/referrals> — keep the path and host exactly as written so any shared mock or gateway matches every call.
- search and q are interchangeable; either query parameter filters by name or service name. Send whichever the candidate's UI logic produces, but only one is needed.
- sort accepts asc or desc; if omitted, the API defaults to desc (newest first), matching the UI's default sort option.
- search/q and sort can be combined in the same request (e.g. ?search=pm&sort=asc) to filter and sort together.
- The API does not page the response — every call above returns the complete matching set. Pagination (10 rows per page) must be implemented client-side on whatever array the API returns.
- Referral detail: GET the same base URL with ?id=<referralId> and the same Authorization header.

Response Parsing

```
{
  "success": true,
  "data": {
    "metrics": [{ "id": "...", "label": "...", "value": "..." }],
    "serviceSummary": {
      "service": "...",
      "yourReferrals": "...",
      "activeReferrals": "...",
      "totalRefEarnings": "..."
    },
    "referral": { "link": "...", "code": "..." },
    "referrals": [
      { "id": 1, "name": "...", "serviceName": "...", "date": "YYYY-MM-DD", "profit": 12345 }
    ]
  }
}
```

The parser must also work if metrics, serviceSummary, and referral sit beside referrals on the same object, as in the reference implementation.

Not Found Page

- Displays a clean "404 - Page Not Found" message
- Includes a primary action link (Back to dashboard) redirecting the user back to "/"

7. Resources & Credentials

Design Resources

- **Login Route:**
https://nxtwave-website-media-files.s3.ap-south-1.amazonaws.com/placement-happenings/webcoding-gifApril14/screenshot-localhost-3000-login-2026-05-26-09_45_54.png
- **Home Route:**
https://nxtwave-website-media-files.s3.ap-south-1.amazonaws.com/placement-happenings/webcoding-gifApril14/screenshot-localhost-3000-2026-05-26-09_46_14.png
- **Referral Detail Route:**
https://nxtwave-website-media-files.s3.ap-south-1.amazonaws.com/placement-happenings/webcoding-gifApril14/screenshot-localhost-3000-referral-48-2026-05-26-09_46_32.png
- **Not Found Route:**
https://nxtwave-website-media-files.s3.ap-south-1.amazonaws.com/placement-happenings/webcoding-gifApril14/screenshot-localhost-3000-not-found-2026-05-26-09_46_54.png

Video Reference

<https://nxtwave-website-media-files.s3.ap-south-1.amazonaws.com/placement-happenings/webcoding-gifApril14/academy-assessment-7-vid.webm>

Test Credentials

Email: admin@example.com
Password: admin123

Submission

1. Public GitHub link or zip file with a README.md file
2. Live deploy URL (Vercel)